

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
	(b)	(i)	$v = \frac{h}{m\lambda}$ transposed and used (1) $v = 1.5 \times 10^7 \text{ [m s}^{-1}\text{]} (1)$		2		2	2	
		(ii)	Electrons are wavelike or diffract or interfere (1) Accept reference to wave-particle duality Graphite crystals act as diffraction gratings or electrons diffract <u>between (around) atom[s]</u> (through lattice) or λ comparable with gaps between atoms (1)	2			2		
			Question 7 total	4	5	3	12	5	8